I. Determine and check s is or not natural

We know $p(x)$ satisfies

$$\begin{cases} p(0) = 0 \\ p(1) = (2-x)^3|_{x=1} = 1 \\ p'(1) = ((2-x)^3)'|_{x=1} = -3(2-x)^2|_{x=1} = -3 \\ p''(1) = 6(2-x)|_{x=1} = 6 \end{cases}$$

We assume $p(x) = ax^3 + bx^2 + cx + d$, then

$$\begin{cases} d = 0 \\ a+b+c+d = 1 \\ 3a+2b+c = -3 \\ 6a+2b = 6 \end{cases} \Rightarrow \begin{cases} a = 7 \\ b = -18 \\ c = 12 \\ d = 0 \end{cases}$$

So $p(x) = 7x^3 - 18x^2 + 12x$.

$$s''(0) = -36 \neq 0$$

So $s(x)$ is NOT natural cubic spline.

II. Interpolation of f

II.a

Actually we have condition:

$$\begin{cases} p_i(x_i) = f_i \quad \& \quad p_i(x_{i+1}) = f_{i+1}, i = 1, 2, \cdots, n-1 \\ p_i'(x_{i+1}) = p_{i+1}'(x_{i+1}), i = 1, 2, \cdots, n-2 \end{cases}$$

$3n-4$ conditions in all. To determine s uniquely, by the theory of the rank of matrix we need $(n-1)*3 = 3n-3$ at least. So we need an additional condition to ensure the equation has the unique solution.

II.b

Assume $p_i(x) = a_i(x-x_i)^2 + b_i(x-x_i) + c_i$ by (a) we have

$$p_i(x_i) = f_i \quad \& \quad p_i(x_{i+1}) \quad \& \quad p_i'(x_i) = m_i$$

Then $\forall i = 1, 2, \ldots, n-1$:

$$\begin{cases} p_i(x_i) = c_i = f_i \\ p_i'(x_i) = b_i = m_i \\ p_i(x_{i+1}) = a_i(x_{i+1}-x_i)^2 + b_i(x_{i+1}-x_i) + c_i = f_{i+1} \end{cases} \Rightarrow \begin{cases} a_i = \frac{f_{i+1}-f_i}{(x_{i+1}-x_i)^2} - \frac{m_i}{x_{i+1}-x_i} \\ b_i = m_i \\ c_i = f_i \end{cases}$$

Then $\forall i = 1, 2, \ldots, n-1$, $p_i(x) = (\frac{f_{i+1}-f_i}{(x_{i+1}-x_i)^2} - \frac{m_i}{x_{i+1}-x_i})(x-x_i)^2 + m_i(x-x_i) + f_i$

II.c

By (b) we get

$$m_{i+1} = p_i'(x_{i+1}) = 2\frac{f_{i+1}-f_i}{x_{i+1}-x_i} - m_i$$

So we can compute m_2, then m_3, $\cdots m_{n-1}$ by induction.

III. Determine a natural cubic spline

Assume $s_2(x) = ax^3 + bx^2 + dx + e$. Then we know

$$\begin{cases} s_1''(-1) = 0 \\ s_2''(1) = 0 \\ s_2(0) = s_1(0) = 1 + c \\ s_2'(0) = s_1'(0) = 1 + 3c \\ s_2''(0) = s_1''(0) = 6c \end{cases} \Rightarrow \begin{cases} 0 = 0 \\ 6a + 2b = 0 \\ e = 1 + c \\ d = 1 + 3c \\ 2b = 6c \end{cases} \Rightarrow \begin{cases} a = -c \\ e = 1 + c \\ d = 3c \\ b = 3c \end{cases}$$

So $s_2(x) = -cx^3 + 3cx^2 + 3cx + 1 + c$. Then if $s(1) = -1 \Rightarrow s_2(1) = -c + 3c + 3c + 1 + c = 6c + 1 = -1 \Rightarrow c = -\frac{1}{3}$.

IV. Cubic spline interpolation of given function

IV.a

We know

$$\begin{cases} f(-1) = 0 \\ f(0) = 1 \\ f(1) = 0 \end{cases}$$

Assume

$$s(x) = \begin{cases} p_1(x) = a_1x^3 + b_1x^2 + c_1x + d_1, x \in [-1, 0] \\ p_2(x) = a_2x^3 + b_2x^2 + c_2x + d_2, x \in [0, 1] \end{cases}$$

Then we get

$$\begin{cases} -a_1 + b_1 - c_1 + d_1 = 0 \\ d_1 = d_2 = 1 \\ a_2 + b_2 + c_2 + d_2 = 0 \\ c_1 = c_2 \\ b_1 = b_2 \\ 6a_2 + 2b_2 = 0 \\ -6a_1 + 2b_1 = 0 \end{cases} \Rightarrow \begin{cases} a_1 = -\frac{1}{2} \\ a_2 = \frac{1}{2} \\ b_1 = -\frac{3}{2} \\ b_2 = -\frac{3}{2} \\ c_1 = 0 \\ c_2 = 0 \\ d_1 = 1 \\ d_2 = 1 \end{cases}$$

So

$$s(x) = \begin{cases} p_1(x) = -\frac{1}{2}x^3 - \frac{3}{2}x^2 + 1, x \in [-1, 0] \\ p_2(x) = \frac{1}{2}x^3 - \frac{3}{2}x^2 + 1, x \in [0, 1] \end{cases}$$

IV.b

By Newton's interpolation, we get $g(x) = 1 - x^2$. Then $g''(x) \equiv -2$. so

$$\int_{-1}^{1} |g''(x)|^2 dx = 8 > \int_{-1}^{1} |s''(x)|^2 dx = \int_{0}^{1} |3x - 3|^2 dx + \int_{-1}^{0} |3x + 3|^2 dx = 3 + 3 + 18 - 9 - 9 = 6$$

Then we vertify that natural cubic splines have the minimal total bending energy.

V. B-spline

V.a

By the defination of B-spline

$$B_i^{n+1}(x) = \frac{x - t_{i-1}}{t_{i+n} - t_{i-1}} B_i^n(x) + \frac{t_{i+n+1} - x}{t_{i+n+1} - t_i} B_{i+1}^n(x)$$

$$B_i^0(x) = \begin{cases} 1, x \in (t_{i-1}, t_i] \\ 0 \qquad otherwise \end{cases}$$

By the defination of hat function

$$\hat{B}_i(x) = \begin{cases} \frac{x-t_{i-1}}{t_i-t_{i-1}}, x \in (t_{i-1}, t_i] \\ \frac{t_{i+1}-x}{t_{i+1}-t_i}, x \in (t_i, t_{i+1}] \\ 0 \quad\quad otherwise \end{cases}$$

Then we know $B_i^1(x) = \hat{B}_i(x)$. Then we get

$$B_i^2(x) = \frac{x-t_{i-1}}{t_{i+1}-t_{i-1}} B_i^1(x) + \frac{t_{i+2}-x}{t_{i+2}-t_i} B_{i+1}^1(x) = \begin{cases} \frac{(x-t_{i-1})^2}{(t_i-t_{i-1})(t_{i+1}-t_{i-1})}, x \in (t_{i-1}, t_i] \\ \frac{(x-t_{i-1})(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(x-t_i)(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, x \in (t_i, t_{i+1}] \\ \frac{(x-t_{i+2})^2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, x \in (t_{i+1}, t_{i+2}] \\ 0 \quad\quad otherwise \end{cases}$$

V.b

Notice that

$$\frac{d}{dx} B_i^2(x) = \begin{cases} \frac{2(x-t_{i-1})}{(t_i-t_{i-1})(t_{i+1}-t_{i-1})}, x \in (t_{i-1}, t_i] \\ \frac{(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+1}-t_i)} - \frac{(x-t_{i-1})}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} - \frac{(x-t_i)}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, x \in (t_i, t_{i+1}] \\ \frac{2(x-t_{i+2})}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, x \in (t_{i+1}, t_{i+2}] \\ 0 \quad\quad otherwise \end{cases}$$

We get

$$\frac{d}{dx} B_i^2(t_i^+) = \frac{1}{(t_{i+1}-t_{i-1})} + \frac{1}{(t_{i+1}-t_i)} - \frac{(t_i-t_{i-1})}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} = \frac{2}{(t_{i+1}-t_{i-1})}$$

$$\frac{d}{dx} B_i^2(t_i^-) = \frac{2}{(t_{i+1}-t_{i-1})}$$

$$\frac{d}{dx} B_i^2(t_{i+1}^-) = \frac{(t_{i+2}-t_{i+1})}{(t_{i+2}-t_i)(t_{i+1}-t_i)} - \frac{1}{(t_{i+1}-t_i)} - \frac{1}{(t_{i+2}-t_i)} = -\frac{2}{(t_{i+2}-t_i)}$$

$$\frac{d}{dx} B_i^2(t_{i+1}^+) = -\frac{2}{(t_{i+2}-t_i)}$$

So $\frac{d}{dx} B_i^2(x)$ is continuous at t_i and t_{i+1}.

V.c

Notice that

$$\frac{d^2}{dx^2} B_i^2(x) = \begin{cases} \frac{2}{(t_i-t_{i-1})(t_{i+1}-t_{i-1})}, x \in (t_{i-1}, t_i] \\ \frac{-2}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{-2}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, x \in (t_i, t_{i+1}] \\ \frac{2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, x \in (t_{i+1}, t_{i+2}] \\ 0 \quad\quad otherwise \end{cases}$$

Besides, easy to find that $\frac{d}{dx} B_i^2(x) > 0, x \in (t_{i-1}, t_i]$ and $\frac{d^2}{dx^2} B_i^2(x) < 0, x \in (t_i, t_{i+1}]$, so suffices to show $\exists! x^* \in (t_i, t_{i+1})$, satisfied $\frac{d}{dx} B_i^2(x) = 0$ By (b), we get $\frac{d}{dx} B_i^2(t_i^+) > 0$ and $\frac{d}{dx} B_i^2(t_{i+1}^-) < 0$, so by intermediate value theorem of monotonic function, we know $\exists! x^* \in (t_i, t_{i+1})$, s.t. $\frac{d}{dx} B_i^2(x) = 0$.

$$\frac{(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+1}-t_i)} - \frac{(x-t_{i-1})}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} - \frac{(x-t_i)}{(t_{i+2}-t_i)(t_{i+1}-t_i)} = 0$$

$\iff$

$$\frac{(t_{i+1}+t_{i-1}-2x)}{(t_{i+1}-t_{i-1})} + \frac{(t_{i+2}+t_i-2x)}{(t_{i+2}-t_i)} = 0$$

$\iff$

$$(t_{i+2}-t_i)(t_{i+1}+t_{i-1}-2x) + (t_{i+1}-t_{i-1})(t_{i+2}+t_i-2x) = 0$$

$\iff$

$$(t_{i+2}-t_i)(t_{i+1}+t_{i-1}) + (t_{i+1}-t_{i-1})(t_{i+2}+t_i) = 2x(t_{i+2}+t_{i+1}-t_{i-1}-t_i)$$

$\Rightarrow$

$$x^* = \frac{t_{i+2}t_{i+1}-t_{i-1}t_i}{t_{i+2}+t_{i+1}-t_{i-1}-t_i}$$

V.d

$$B_i^2(x) = \begin{cases} 0 \leqslant \frac{(x-t_{i-1})^2}{(t_i-t_{i-1})(t_{i+1}-t_{i-1})} < \frac{t_i-t_{i-1}}{t_{i+1}-t_{i-1}} < 1, x \in (t_{i-1}, t_i] \\ \frac{(x-t_{i-1})(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(x-t_i)(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, x \in (t_i, t_{i+1}] \\ 0 \leqslant \frac{(x-t_{i+2})^2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})} < \frac{t_{i+2}-t_{i+1}}{t_{i+2}-t_i} < 1, x \in (t_{i+1}, t_{i+2}] \\ 0 \quad otherwise \end{cases}$$

suffices to show

$$\frac{(x-t_{i-1})(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(x-t_i)(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+1}-t_i)} \in [0,1), x \in (t_i, t_{i+1}]$$

By (c) we know $B_i^2(x), x \in (t_i, x^*)$ monotonically increasing, $B_i^2(x), x \in (x^*, t_{i+1})$ monotonically decreasing.

$$0 \leqslant B_i^2(t_i) = \frac{t_i - t_{i-1}}{t_{i+1}-t_{i-1}} < 1$$

$$0 \leqslant B_i^2(t_i) = \frac{t_{i+2} - t_{i+1}}{t_{i+2}-t_i} < 1$$

So suffices to show $B_i^2(x^*) < 1$ i.e.

$$\frac{(x^*-t_{i-1})(t_{i+1}-x^*)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(x^*-t_i)(t_{i+2}-x^*)}{(t_{i+2}-t_i)(t_{i+1}-t_i)}$$

We know

$$x^* = \frac{t_{i+2}t_{i+1} - t_{i-1}t_i}{t_{i+2}+t_{i+1}-t_{i-1}-t_i}.$$

Set

$$D := t_{i+2}+t_{i+1}-t_{i-1}-t_i.$$

$$\begin{aligned} x^* - t_{i-1} &= \frac{t_{i+2}t_{i+1} - t_{i-1}t_i}{D} - t_{i-1} = \frac{t_{i+2}t_{i+1} - t_{i-1}t_i - t_{i-1}D}{D} = \frac{(t_{i+1}-t_{i-1})(t_{i+2}-t_{i-1})}{D}, \\ t_{i+1} - x^* &= t_{i+1} - \frac{t_{i+2}t_{i+1} - t_{i-1}t_i}{D} = \frac{t_{i+1}D - (t_{i+2}t_{i+1} - t_{i-1}t_i)}{D} = \frac{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)}{D}, \\ x^* - t_i &= \frac{t_{i+2}t_{i+1} - t_{i-1}t_i}{D} - t_i = \frac{t_{i+2}t_{i+1} - t_{i-1}t_i - t_iD}{D} = \frac{(t_{i+1}-t_i)(t_{i+2}-t_i)}{D} \\ t_{i+2} - x^* &= t_{i+2} - \frac{t_{i+2}t_{i+1} - t_{i-1}t_i}{D} = \frac{t_{i+2}D - (t_{i+2}t_{i+1} - t_{i-1}t_i)}{D} = \frac{(t_{i+2}-t_{i-1})(t_{i+2}-t_i)}{D}. \end{aligned}$$

Then

$$B_i^2(x^*) = \frac{\frac{(t_{i+1}-t_{i-1})(t_{i+2}-t_{i-1})}{D} \cdot \frac{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)}{D}}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{\frac{(t_{i+1}-t_i)(t_{i+2}-t_i)}{D} \cdot \frac{(t_{i+2}-t_{i-1})(t_{i+2}-t_i)}{D}}{(t_{i+2}-t_i)(t_{i+1}-t_i)}$$

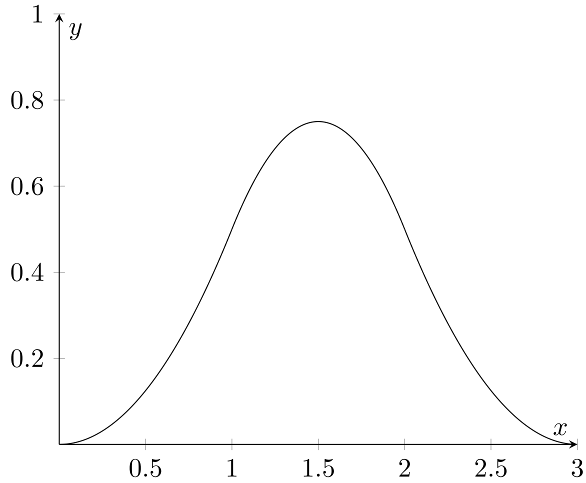
$$= \frac{(t_{i+1}-t_{i-1})(t_{i+2}-t_{i-1})}{D^2} + \frac{(t_{i+2}-t_{i-1})(t_{i+2}-t_i)}{D^2}$$

$$= \frac{t_{i+2}-t_{i-1}}{D^2}\Big[(t_{i+1}-t_{i-1}) + (t_{i+2}-t_i)\Big] = \frac{t_{i+2}-t_{i-1}}{D^2}\big(t_{i+1}+t_{i+2}-t_{i-1}-t_i\big) = \frac{t_{i+2}-t_{i-1}}{D}.$$

So $B_i^2(x^*) = \frac{t_{i+2}-t_{i-1}}{t_{i+2}+t_{i+1}-t_{i-1}-t_i} < 1$.

V.e

We take $i = 1$ situation as an example. In general cases, only need to replace 0 with $i-1$, 1 with i and so on.



VI. Hermite Interpolation of given points

VI.a

$f[0] = 1, f[1] = 2, f[3] = 0$
$f[0,1] = 1, f[1,1] = -1, f[1,3] = -1, f[3,3] = 0$
$f[0,1,1] = -2, f[1,1,3] = 0, f[1,3,3] = \frac{1}{2}$
$f[0,1,1,3] = \frac{2}{3}, f[1,1,3,3] = \frac{1}{4}$
$f[0,1,1,3,3] = -\frac{5}{36}$
So,$f(2) = 1 + (2-0) - 2(2-0)(2-1) + \frac{2}{3}(2-0)(2-1)^2 - \frac{5}{36}(2-0)(2-1)^2(2-3) = 1 + 2 - 4 + \frac{4}{3} + \frac{5}{18} = \frac{11}{18}$

VI.b

By Hermite Interpolation, we know

$$f(x) - p(f;x) = f[0,1,1,3,3,x]x(x-1)^2(x-3)^2 = \frac{f^{(5)}(\xi(x))}{5!}x(x-1)^2(x-3)^2$$

Considering $x(x-1)^2(x-3)^2_{\max} \approx 2.0594$ So the error

$$|f(x) - p(f;x)| \leqslant 2.06\frac{M}{5!}$$

VII. Prove equation

We prove by induction, when $k = 1$, $\Delta f(x) = hf[x_0, x_1]$, $\nabla f(x) = hf[x_0, x_{-1}]$.
Assume that $k = n$ has been proved, when $k = n + 1$:

$$\Delta^{k+1} f(x) = \Delta^k f(x+h) - \Delta^k f(x) = k!h^k f[x_1, x_2, \ldots, x_{k+1}] - k!h^k f[x_0, x_2, \ldots, x_k]$$

then By Newton Interpolation:

$$\Delta^{k+1} f(x) = (k+1)!h^{k+1} f[x_0, x_1, \ldots, x_{k+1}]$$

the same way for ∇ Assume that $k = n$ has been proved, when $k = n + 1$:

$$\nabla^{k+1} f(x) = \nabla^k f(x) - \nabla^k f(x-h) = k!h^k f[x_0, x_{-1}, \ldots, x_{-k}] - k!h^k f[x_{-1}, x_{-2}, \ldots, x_{-k-1}]$$

then By Newton Interpolation:

$$\nabla^{k+1} f(x) = (k+1)!h^{k+1} f[x_0, x_{-1}, \ldots, x_{-k-1}]$$

VIII. Prove equation

By defination of $\frac{\partial f}{\partial x_0}$,we know

$$\frac{\partial f[x_0, x_1, \dots, x_n]}{\partial x_0} = \lim_{\Delta x \to 0} \frac{f[x_0 + \Delta x, x_1, \dots, x_n] - f[x_0, x_1, \dots, x_n]}{\Delta x} = \lim_{\Delta x \to 0} f[x_0 + \Delta x, x_0, x_1, \dots, x_n]$$

And f is differentiable at x_0, so f is continuous at x_0, so with finite steps, the error simulate in progress will control by a const c and ε by the continuous at x_0.i.e.

$$|f[x_0 + \Delta x] - f[x_0]| < \varepsilon, |f[x_0 + \Delta x, x_0] - f[x_0, x_0]| < \varepsilon$$

we can prove by induction,every interpolation has only a $c\varepsilon$ difference, in which c is a constant $\Rightarrow$

$$\lim_{\Delta x \to 0} f[x_0 + \Delta x, x_0, x_1, \dots, x_n] = f[x_0, x_0, x_1, \dots, x_n]$$

IX. Min-Max problem

i.e. to solve:

$$\min_{a_i \in \mathbb{R}, i \in 1,2,\cdots n} \max_{x \in [a,b]} |a_0 x^n + a_1 x^{n-1} + \cdots + a_n|$$